RBC with Endogenous Labor

Chengyuan He

April 2026

1 HP filter

HP filter: decompose economic series into trend and cyclical component

$$y_t = y_t^c + y_t^T$$

$$\min_{\{y_t^T\}_{t=1}^N} \mathcal{L} = \underbrace{\sum_{t=1}^{N} (y_t - y_t^T)^2}_{(1)} + \lambda \underbrace{\sum_{t=2}^{N-1} \left[(y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T) \right]^2}_{(2)}$$

Notes:

1) penalty of calling everything trend

2) change in trend growth rate, penalty of wiggles

If λ huge, close to straight line in logs.
If $\lambda = 0$, wiggle a lot, fit to y_t.
Define

$$\Delta^2 y_t^T \equiv y_{t+1}^T - 2y_t^T + y_{t-1}^T \qquad \text{(second-order difference)}$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = 0 \quad \Rightarrow \quad \frac{\partial (1)}{\partial y_t^T} = 2(y_t^T - y_t)$$

Since

$$\begin{aligned} \Delta^2 y_{t-1}^T &= y_t^T - 2y_{t-1}^T + y_{t-2}^T && \text{(coefficient +1)} \\ \Delta^2 y_t^T &= y_{t+1}^T - 2y_t^T + y_{t-1}^T && \text{(coefficient −2)} \\ \Delta^2 y_{t+1}^T &= y_{t+2}^T - 2y_{t+1}^T + y_t^T && \text{(coefficient +1)} \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 &= 2\Delta^2 y_{t-1}^T \frac{\partial \Delta^2 y_{t-1}^T}{\partial y_t^T} + 2\Delta^2 y_t^T \frac{\partial \Delta^2 y_t^T}{\partial y_t^T} + 2\Delta^2 y_{t+1}^T \frac{\partial \Delta^2 y_{t+1}^T}{\partial y_t^T} \\ &= 2\big(\Delta^2 y_{t-1}^T - 2\Delta^2 y_t^T + \Delta^2 y_{t+1}^T\big) \end{aligned}$$

Since

$$\Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T,$$

$$-2\Delta^2 y_t^T = -2(y_{t+1}^T - 2y_t^T + y_{t-1}^T) = -2y_{t+1}^T + 4y_t^T - 2y_{t-1}^T,$$

$$\Delta^2 y^T_{t+1} = y^T_{t+2} - 2y^T_{t+1} + y^T_t,$$

we get

$$\frac{\partial \mathcal{L}}{\partial y^T_t} = \frac{\partial(1)}{\partial y^T_t} + \lambda \frac{\partial}{\partial y^T_t} \sum_{s=2}^{N-1} (\Delta^2 y^T_s)^2 = 0.$$

$$\Rightarrow \quad (1+6\lambda)y^T_t - 4\lambda y^T_{t-1} - 4\lambda y^T_{t+1} + \lambda y^T_{t-2} + \lambda y^T_{t+2} = y_t, \qquad 3 \le t \le N-2.$$

At $t = 1$ (only $\Delta^2 y^T_2$ exists),

$$(1+\lambda)y^T_1 - 2\lambda y^T_2 + \lambda y^T_3 = y_1.$$

At $t = 2$ (only $\Delta^2 y^T_2$ and $\Delta^2 y^T_3$ exist),

$$-2\lambda y^T_1 + (1+5\lambda)y^T_2 - 4\lambda y^T_3 + \lambda y^T_4 = y_2.$$

Symmetrically for $t = N$ and $t = N-1$, collect $\{y_t\}$ into the $N \times 1$ vector y and $\{y^T_t\}$ into y^T, then we have

$$Ay^T = y \quad \Rightarrow \quad y^T = A^{-1}y \quad \Rightarrow \quad y^c = y - A^{-1}y.$$

where

$$A = \begin{pmatrix} 1+\lambda & -2\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 \\ -2\lambda & 1+5\lambda & -4\lambda & \lambda & 0 & \cdots & 0 & 0 & 0 \\ \lambda & -4\lambda & 1+6\lambda & -4\lambda & \lambda & \cdots & 0 & 0 & 0 \\ 0 & \lambda & -4\lambda & 1+6\lambda & -4\lambda & \ddots & 0 & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -4\lambda & 1+6\lambda & -4\lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda & -4\lambda & 1+5\lambda & -2\lambda \\ 0 & 0 & 0 & \cdots & 0 & 0 & \lambda & -2\lambda & 1+\lambda \end{pmatrix}.$$

Let D be the $(N-2) \times N$ second-difference matrix, whose t-th row ($t = 2, \ldots, N-1$) looks like

$$[\, 0 \ \cdots \ 1 \ -2 \ 1 \ \cdots \ 0 \,]$$

so that

$$(Dy^T)_{t-1} = \Delta^2 y^T_t.$$

The HP objective becomes

$$\min_{y^T} \, (y - y^T)'(y - y^T) + \lambda (Dy^T)'(Dy^T).$$

F.O.C.

$$-2(y - y^T) + 2\lambda D'Dy^T = 0 \quad \Rightarrow \quad (I + \lambda D'D)y^T = y.$$

Hence

$$A = I + \lambda D'D.$$

As a 5-diagonal pattern:

$$\text{interior diagonal: } [\, 0 \ \cdots \ 1 \ -2 \ 1 \ \cdots \ 0 \,] \begin{bmatrix} 0 \\ \vdots \\ 1 \\ -2 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 6$$

$$\text{first-off diagonal: } [\,0\ \cdots\ 1\ -2\ 1\ 0\ \cdots\ 0\,] \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = -4$$

$$\text{second-off diagonal: } [\,0\ \cdots\ 1\ -2\ 1\ 0\ 0\ \cdots\ 0\,] \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1$$

$$\lambda \uparrow \quad \Rightarrow \quad \text{the smoother the series } y_t^T.$$

Pros

1) Simple and transparent. (defined by one objective)
2) Produces a smooth trend with tunable smoothness λ
3) Widely used in Macro

Cons

1) End-point problem: start and end of sample is distorted.
2) λ sensitivity: moments (volatility, correlations) change a lot with λ.
3) Spurious dynamics: HP can induce artificial serial correlations and distort cross-corr between series (create cycles).
4) Leakage: low-frequency cycles leak into trend.
5) Not structural: no model interpretation.

Why $\lambda = 1600$ is the quarterly standard?
Rule of thumb:

$$\lambda = 100 \times (\text{periods per year})^2.$$

Hence

$$\lambda_{\text{quarter}} = 100 \times 4^2 = 1600, \qquad \lambda_{\text{month}} = 100 \times 12^2 = 14400, \qquad \lambda_{\text{annual}} = 100 \times 1^2 = 100.$$

2 Long and Plosser (1983, JPE)

Social planner's Problem Setup:

$$U = \max_{\{C_t, L_t, K_t\}} E_0 \left\{ \sum_{t=0}^{\infty} \beta^t \left[\theta \ln C_t + (1-\theta) \ln L_t \right] \right\}$$

s.t.

$$K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - C_t \qquad \text{and } K_0 \text{ is given.}$$

Resource constraint:

$$C_t + K_t = Y_t.$$

$$N_t + L_t = 1 \qquad \text{(trade-off)}$$

$$L: \text{ leisure} \qquad N: \text{ working hour.}$$

Total time is normalized to 1.

where

$$\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{i.i.d.}{\sim} N(0, \sigma^2), \qquad \psi < 1.$$

Note: capital completely turns over,

$$(1-\delta) K_{t-1} = 0 \qquad \delta = 1.$$

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big[\theta \ln C_t + (1-\theta) \ln(1-N_t) + \lambda_t \big(Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - C_t - K_t \big) \Big].$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \frac{\theta}{C_t} = \lambda_t \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \qquad \Rightarrow \qquad -\underbrace{\frac{1-\theta}{1-N_t}}_{\text{utility loss}} + \underbrace{\lambda_t}_{\text{MUC}_t} \underbrace{(1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}}_{\text{MPN}_t} = 0$$

$$\Rightarrow \quad \frac{1-\theta}{1-N_t} = (1-\alpha) \lambda_t Z_t K_{t-1}^{\alpha} N_t^{-\alpha} \tag{2}$$

labor-leisure condition (intratemporal)

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t \left[\lambda_{t+1} \frac{\partial}{\partial K_t} \big(Z_{t+1} K_t^{\alpha} N_{t+1}^{1-\alpha} \big) \right] = 0$$

$$\Rightarrow \quad \lambda_t = \beta E_t \big(\lambda_{t+1} \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} \big) \tag{3}$$

Euler equation (intertemporal)

TVC:

$$\lim_{T \to \infty} \beta^T U_C(C, L) K_{T+1} = 0. \tag{4}$$

Closed-form solution: constant savings rate and constant labor
Tractability: log utility + Cobb–Douglas + full depreciation

$$\Rightarrow \quad \text{constant savings rate / hours.}$$

Since

$$Y_{t+1} = Z_{t+1} K_t^{\alpha} N_{t+1}^{1-\alpha},$$

we have

$$\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} = \alpha \frac{Y_{t+1}}{K_t}.$$

Plug into (3):

$$\lambda_t = \beta E_t \left(\lambda_{t+1} \alpha \frac{Y_{t+1}}{K_t} \right).$$

Using (1):

$$\frac{\theta}{C_t} = \beta E_t \left(\frac{\theta}{C_{t+1}} \alpha \frac{Y_{t+1}}{K_t} \right)$$
$$\Rightarrow \quad \frac{1}{C_t} = \beta \alpha E_t \left[\frac{Y_{t+1}}{C_{t+1}} \frac{1}{K_t} \right]. \tag{5}$$

Guess and verify:

$$C_t = (1-s) Y_t, \qquad s \text{ is the savings rate}, \qquad K_t = s Y_t. \qquad \text{(full depreciate)}$$

Then

$$\frac{Y_{t+1}}{C_{t+1}} = \frac{1}{1-s}.$$

Plug into (5):

$$\frac{1}{(1-s) Y_t} = \alpha\beta \frac{1}{1-s} \frac{1}{s Y_t}$$
$$\Rightarrow \quad s = \alpha\beta.$$

Hence

$$C_t = (1-\alpha\beta) Y_t, \qquad K_t = \alpha\beta \, Y_t.$$

$$\begin{cases} \alpha \uparrow \Rightarrow \text{ capital income share} \uparrow \Rightarrow \text{ savings rate} \uparrow, \\ \beta \uparrow \Rightarrow \text{ discount factor} \downarrow \Rightarrow \text{ savings rate} \uparrow. \end{cases}$$

Plug into (2):

$$\frac{1-\theta}{1-N_t} = (1-\alpha) \frac{\theta}{C_t} \frac{Y_t}{N_t} = (1-\alpha) \frac{\theta}{(1-\alpha\beta) Y_t} \frac{Y_t}{N_t}$$
$$\Rightarrow \quad (1-\theta)(1-\alpha\beta) N_t = (1-\alpha)\theta(1-N_t)$$
$$\Rightarrow \quad N_t = \bar{N} = \frac{\theta(1-\alpha)}{\theta(1-\alpha) + (1-\theta)(1-\alpha\beta)}.$$

Since N_t = constant $\forall t$, we have

$$\hat{n}_t = 0.$$

Also

$$K_t = \alpha\beta Y_t, \qquad C_t = (1-\alpha\beta)Y_t$$

so

$$\hat{k}_t = \hat{c}_t = \hat{y}_t.$$

Since

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1} + (1-\alpha)\hat{n}_t = \hat{z}_t + \alpha\hat{k}_{t-1},$$

we get

$$\hat{k}_t = \hat{c}_t = \hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1}.$$

And

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

TFP shock alone can generate output/consumption/investment fluctuations, but

1) labor does not fluctuate at all,

2) c, k, y move one-for-one.

Consumption is usually smoother than output; capital is usually more volatile.

3) Labor productivity moves like output in model.

$$\left(\frac{\hat{Y}}{\hat{N}}\right)_t = \hat{y}_t - \hat{n}_t = \hat{y}_t, \qquad \text{while in data,}$$

$$\sigma(Y/N) \ll \sigma(Y), \qquad \text{and } \mathrm{Corr}(Y/N, Y) = 0.55.$$

4) Prices (MRS, MRT, MPL, MPK) too-tight with output & TFP, hard to generate weak wage cyclicality & interest rate counter-cyclicality.

Labor market equilibrium
Household's problem:

$$U_{N,t} + \lambda_t W_t = 0 \quad \Rightarrow \quad U_{N,t} + U_{C,t} W_t = 0$$

labor supply
Firm's problem:

$$\max_{K_{t-1}, N_t} \ \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

labor demand
Hence

$$W_t = Z_t \frac{Y_t}{N_t}(1-\alpha).$$

From household labor supply:

$$W_t = \frac{1-\theta}{\theta}\frac{C_t}{1-N_t}.$$

Equivalently,

$$W_t = \frac{1-\theta}{\theta} \frac{(1-\alpha\beta) Z_t K_{t-1}^{\alpha} N_t^{1-\alpha}}{1-N_t}.$$

From firm labor demand:

$$W_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}.$$

A positive Z_t or K_{t-1} shock shifts labor supply up by the same factor as labor demand:

$$Z_t K_{t-1}^{\alpha}.$$

Hence labor supply's wealth effect is exactly offset by labor demand increase. So when Z_t rises, the real wage increases. This generates a substitution effect toward more labor, but also an income effect toward more leisure.

$$W_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}$$

$$Z_t \uparrow \Rightarrow \; MPN_t \uparrow \Rightarrow \; W_t \uparrow \Rightarrow \; \text{labor demand curve shifts up}$$

$$Z_t \uparrow \Rightarrow \; Y_t \uparrow \Rightarrow \; C_t = (1-\alpha\beta) Y_t \uparrow \Rightarrow \; MU_{C,t} \downarrow \Rightarrow \; N_t \downarrow \qquad \text{Wealth effect}$$

$$\underbrace{\frac{1-\theta}{1-N_t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\text{marginal utility value of wage income}} = \underbrace{MU_{C,t} W_t}_{\text{intra-temporal optimality}} = \frac{\theta}{C_t}(1-\alpha)\frac{Y_t}{N_t}$$

$$= \frac{\theta(1-\alpha)}{N_t}\left(\frac{Y_t}{C_t}\right) = \frac{\theta(1-\alpha)}{1-\alpha\beta}\frac{1}{N_t}.$$

Hansen (1985): Basic idea Expand log preference to make labor respond to shocks.
Household's problem:

$$\max_{\{C_t, L_t, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) = E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$$

$$\text{s.t.} \qquad N_t + L_t = 1$$

$$C_t + K_t = W_t N_t + R_t K_{t-1} + (1-\delta) K_{t-1} \qquad \text{(budget constraint not resource constraint)}$$

$$K_t - (1-\delta) K_{t-1} = I_t \qquad \text{investment}$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big[u(C_t, 1 - N_t) + \lambda_t \big(W_t N_t + R_t K_{t-1} + (1-\delta) K_{t-1} - C_t - K_t \big) \Big]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \underbrace{U_{C,t}}_{\text{marginal utility of consumption}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N,t} + \lambda_t W_t = 0$$

$$\Rightarrow \quad - \underbrace{U_{N,t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\substack{\text{marginal utility value of} \\ \text{one more unit of wage income}}}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t[\lambda_{t+1} R_{t+1} + \lambda_{t+1}(1-\delta)] = 0$$

$$\Rightarrow \quad U_{C,t} = \beta E_t [U_{C,t+1}(R_{t+1} + 1 - \delta)]$$

Firm's problem

$$\max_{N_t, K_{t-1}} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

s.t.

$$Y_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha}, \qquad \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t.$$

FOCs:

$$\frac{\partial \Pi_t}{\partial K_{t-1}} = 0 \quad \Rightarrow \quad \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} = R_t \quad \Rightarrow \quad \alpha \frac{Y_t}{K_{t-1}} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t} = 0 \quad \Rightarrow \quad (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha} = W_t \quad \Rightarrow \quad (1-\alpha) \frac{Y_t}{N_t} = W_t.$$

Let utility function follow

$$U(C_t, 1 - N_t) = \ln C_t + A \ln(1 - N_t).$$

Then

$$U_{C,t} = \frac{1}{C_t}, \qquad U_{N,t} = -\frac{A}{1 - N_t}.$$

Hence equilibrium conditions are

$$\begin{cases} 1 = \beta E_t \left[\dfrac{C_t}{C_{t+1}} (R_{t+1} + 1 - \delta) \right], \\ \dfrac{A}{1 - N_t} = \dfrac{1}{C_t} W_t, \\ W_t = (1 - \alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \\ R_t = \alpha Z_t K_{t-1}^{\alpha - 1} N_t^{1-\alpha}, \\ \Pi_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} - W_t N_t - R_t K_{t-1} = 0, \\ C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1}, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Do not substitute out R_t, W_t, to make a system of equations of (C_t, N_t, K_t, Z_t).
Substituting yields

$$\begin{cases} 1 = \beta E_t \left[\dfrac{C_t}{C_{t+1}} \left(\alpha Z_{t+1} K_t^{\alpha - 1} N_{t+1}^{1-\alpha} + 1 - \delta \right) \right], \\ A C_t = (1 - N_t)(1 - \alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \\ K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Step 1: Steady state $\Rightarrow$ Step 2: Log-linearize $\Rightarrow$ Step 3: Numerically solve

3 Steady state

At steady state:

$$K_t = K_{t-1} = \bar{K}, \qquad C_t = C_{t+1} = \bar{C}, \qquad N_t = N_{t+1} = \bar{N}.$$

$$\bar{K} = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha} + (1-\delta)\bar{K} - \bar{C} \quad \Rightarrow \quad \bar{C} = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha} - \delta\bar{K}. \tag{1}$$

$$1 = \alpha\beta\bar{Z}\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} + (1-\delta)\beta. \tag{2}$$

$$A\bar{C} = (1-\bar{N})(1-\alpha)\bar{Z}\bar{K}^{\alpha}\bar{N}^{-\alpha}. \tag{3}$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha}\bar{N}^{1-\alpha} - \delta\bar{K} = \frac{1}{A}\Big((1-\alpha)\bar{Z}\,\bar{K}^{\alpha}\bar{N}^{-\alpha} - (1-\alpha)\bar{Z}\,\bar{K}^{\alpha}\bar{N}^{1-\alpha}\Big)$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} + \frac{1-\alpha}{A}\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} = \frac{1-\alpha}{A}\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{-\alpha} + \delta$$

$$\Rightarrow \quad \bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{1-\alpha} = \frac{A}{1-\alpha+A}\left[\frac{1-\alpha}{A}\bar{Z}\,\bar{K}^{\alpha-1}\bar{N}^{-\alpha} + \delta\right]$$

Using (2), after simplification one gets

$$\bar{N} = \left\{1 + \frac{A}{1-\alpha}\left[1 - \frac{\alpha\beta\delta}{1-\beta(1-\delta)}\right]\right\}^{-1}.$$

Then

$$\bar{K} = \left[\frac{1-\beta(1-\delta)}{\bar{Z}\alpha\beta}\bar{N}^{\alpha-1}\right]^{\frac{1}{\alpha-1}} = \bar{N}\left[\frac{\bar{Z}\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{1}{1-\alpha}}.$$

And

$$\bar{C} = \bar{Z}\,\bar{K}^{\alpha}\bar{N}^{\,1-\alpha} - \delta\bar{K} = \bar{N}^{\alpha}\cdot\left[\frac{\bar{Z}\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{\alpha}{1-\alpha}}\bar{N}^{1-\alpha}\bar{Z} - \delta\bar{K}$$

$$= \bar{Z}^{\frac{1}{1-\alpha}}\bar{N}\left[\frac{\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{\alpha}{1-\alpha}} - \delta\bar{N}\,\bar{Z}^{\frac{1}{1-\alpha}}\left[\frac{\alpha}{\beta^{-1}-(1-\delta)}\right]^{\frac{1}{1-\alpha}}$$

$$= \bar{N}\,\bar{Z}^{\frac{1}{1-\alpha}}\left[\left(\frac{\alpha}{\beta^{-1}-(1-\delta)}\right)^{\frac{\alpha}{1-\alpha}} - \delta\left(\frac{\alpha}{\beta^{-1}-(1-\delta)}\right)^{\frac{1}{1-\alpha}}\right]$$

Derive the log-linearized system
(i) TFP process:

$$\log Z_t = (1-\psi)\log\bar{Z} + \psi\log Z_{t-1} + \varepsilon_t$$

Let

$$Z_t = \bar{Z}e^{\hat{z}_t}.$$

Then

$$\hat{z}_t + \log\bar{Z} = (1-\psi)\log\bar{Z} + \psi\log\bar{Z} + \psi\hat{z}_{t-1} + \varepsilon_t$$

$$\Rightarrow \quad \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

(ii)

$$K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta)K_{t-1} - C_t.$$

Substitute

$$K_t = \bar{K}(1+\hat{k}_t), \quad Z_t = \bar{Z}(1+\hat{z}_t), \quad K_{t-1} = \bar{K}(1+\hat{k}_{t-1}), \quad N_t = \bar{N}(1+\hat{n}_t), \quad C_t = \bar{C}(1+\hat{c}_t),$$

and keep only first-order terms:

$$\bar{K} + \bar{K}\hat{k}_t = \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha}(1+\hat{z}_t)(1+\alpha\hat{k}_{t-1})(1+(1-\alpha)\hat{n}_t) + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C} - \bar{C}\hat{c}_t.$$

Using the steady state condition and dropping higher-order terms:

$$\bar{K}\hat{k}_t = \alpha\bar{Y}\hat{k}_{t-1} + \bar{Y}\hat{z}_t + (1-\alpha)\bar{Y}\hat{n}_t + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C}\hat{c}_t,$$

where

$$\bar{Y} \equiv \bar{Z}\bar{K}^{\alpha}\bar{N}^{1-\alpha}.$$

Hence

$$\hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + (1-\alpha)\frac{\bar{Y}}{\bar{K}}\hat{n}_t + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t.$$

(iii)

$$AC_t = (1-N_t)(1-\alpha)Z_t K_{t-1}^{\alpha} N_t^{-\alpha}.$$

Log-linearizing gives

$$\hat{n}_t = \left(\alpha + \frac{\bar{N}}{1-\bar{N}}\right)^{-1}\left(\hat{z}_t + \alpha\hat{k}_{t-1} - \hat{c}_t\right).$$

(iv) Euler equation:

$$1 = \beta E_t\left[\frac{C_t}{C_{t+1}}\left(\alpha Z_{t+1}K_t^{\alpha-1}N_{t+1}^{1-\alpha} + 1 - \delta\right)\right].$$

Let

$$M_{t+1} \equiv \alpha Z_{t+1}K_t^{\alpha-1}N_{t+1}^{1-\alpha}, \qquad G_{t+1} \equiv M_{t+1} + 1 - \delta.$$

Then

$$\hat{m}_{t+1} = \hat{z}_{t+1} + (\alpha-1)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}.$$

Define

$$\tilde{\delta} \equiv \frac{\bar{M}}{\bar{G}}.$$

Since

$$\bar{G} = \beta^{-1}, \qquad \bar{G} = \bar{M} + (1-\delta),$$

we have

$$\tilde{\delta} = \frac{\bar{M}}{\bar{G}} = 1 - \frac{1-\delta}{\bar{G}} = 1 - \beta(1-\delta).$$

Thus

$$G_{t+1} \approx \bar{G}(1 + \tilde{\delta}\hat{m}_{t+1}).$$

Also

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C}e^{\hat{c}_t}}{\bar{C}e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}} \approx 1 + \hat{c}_t - \hat{c}_{t+1}.$$

So Euler equation becomes

$$1 = \beta E_t\left[\bar{G}(1 + \hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1})\right]$$

Since

$$\beta\bar{G} = 1,$$

we obtain

$$E_t\left[\hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1}\right] = 0.$$

Hence

$$\hat{c}_t = E_t\hat{c}_{t+1} - \tilde{\delta}E_t\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}\right].$$

If you have CRRA,

$$\frac{C_t}{C_{t+1}} \to \left(\frac{C_t}{C_{t+1}}\right)^{\sigma} \quad \Rightarrow \quad \sigma\hat{c}_t = \sigma E_t\hat{c}_{t+1} - E_t\hat{r}_{t+1}.$$

Thus we have a system of equations:

$$\begin{cases} \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t, \\ \hat{c}_t = E_t\hat{c}_{t+1} - \tilde{\delta}E_t\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t + (1-\alpha)\hat{n}_{t+1}\right], \\ \hat{n}_t = \left(\alpha + \dfrac{\bar{N}}{1-\bar{N}}\right)^{-1}(\hat{z}_t + \alpha\hat{k}_{t-1} - \hat{c}_t), \\ \hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + (1-\alpha)\frac{\bar{Y}}{\bar{K}}\hat{n}_t + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t. \end{cases}$$

Four equations, four variables:

$$(\hat{k}_{t-1}, \hat{c}_t, \hat{n}_t, \hat{z}_t).$$

$\hat{k}_{t-1}$ is predetermined variable, $\hat{z}_t$ follows a stable AR(1) process,
$(\hat{c}_t, \hat{n}_t)$ are jump variables.

Use undetermined coefficient method; guess

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \qquad \hat{n}_t = V_{nk}\hat{k}_{t-1} + V_{nz}\hat{z}_t.$$

Then

$$\hat{k}_t = V_{kz}\sum_{i=0}^{\infty}\sum_{j=0}^{i}V_{kk}^{j}\psi^{i-j}\varepsilon_{t-i}, \qquad \hat{z}_t = \sum_{s=0}^{\infty}\psi^{s}\varepsilon_{t-s},$$

and since

$$\hat{y}_t = \alpha\hat{k}_{t-1} + (1-\alpha)\hat{n}_t + \hat{z}_t,$$

we can write

$$\hat{y}_t = V_{yz}\varepsilon_t + \sum_{i=0}^{\infty}\left(V_{yk}V_{kz}\sum_{j=0}^{i}V_{kk}^{j}\psi^{i-j} + V_{yz}\psi^{i+1}\right)\varepsilon_{t-i-1}.$$

Therefore,

$$\mathrm{Var}(\hat{y}_t) = \left(V_{yz}^2 + \sum_{i=0}^{\infty} \left[V_{yk} V_{kz} \sum_{j=0}^{i} V_{kk}^{j} \psi^{\,i-j} + V_{yz} \psi^{i+1} \right]^2 \right) \mathrm{Var}(\varepsilon_t).$$

Given parameters' value, we can back out the explicit numerical solution.
This model works:

1) well for consumption volatility

2) bad for output volatility: requires too small TFP volatility

3) bad for hours volatility $\Rightarrow$ way too low.

4) bad for investment volatility $\Rightarrow$ far too low.

But it allows N_t to move with shocks compared to Long–Plosser. While with this set of preference and technology, it cannot generate enough labor and investment volatility relative to output.

4 Hansen (1985) with indivisible labor

Labor market phenomena:

1) Hours fluctuate a lot relative to productivity.

2) Micro evidence does not support huge intertemporal substitution of labor at individual level.

Target: get big aggregate hours fluctuations without assuming each person is very willing to shift leisure across time.

Key mechanism:

1) Each individual's labor is a corner-choice:

$$\tilde{n}_t \in \{0, n^*\}.$$

Either work a fixed shift or no work.

2) Lottery over jobs: assigned a job with probability π_t.

3) Complete unemployment insurance: C_t is independent of job status.

Crucial: we can write expected utility clearly at the "representative agent" level.

Thus total labor supply:

$$H_t = \bar{L}\pi_t n^*$$

where $\bar{L}$ is total labor force.

Average labor supply:

$$n_t = \frac{H_t}{\bar{L}} = \pi_t n^*.$$

Representative-agent utility: individual utility is

$$U(C_t) + v(\tilde{n}_t).$$

Given lottery,

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t v(n^*) + (1 - \pi_t)v(0).$$

Let

$$v(0) = 0, \qquad \frac{v(n^*)}{n^*} = -A < 0.$$

Then

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t(-An^*) = U(C_t) - An_t.$$

Suppose

$$U(C_t) = \ln C_t \quad \Rightarrow \quad U_t = \ln C_t - An_t.$$

Hence

$$U_{C,t} = \frac{1}{C_t}, \qquad U_{N,t} = -A, \qquad -U_{N,t} = U_{C,t}W_t$$

Micro-level: no smoothing in labor, either n^* or 0, keep small intertemporal labor substitution.

Macro-level: planner / representative agent can vary π_t a lot, aggregate hours become very responsive.

Thus, we can define the equilibrium conditions for this model:

$$
\begin{cases}
1 = \beta E_t \left[\dfrac{C_t}{C_{t+1}} \left(\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta \right) \right], \\
K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1} - C_t, \\
A C_t = (1-\alpha) Z_t K_{t-1}^{\alpha} N_t^{-\alpha}, \qquad \text{key difference} \qquad -U_{N,t} = U_{C,t} W_t = U_{C,t} MPL_t, \\
\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t.
\end{cases}
$$

Mechanism
In the indivisible labor model, aggregate hours fluctuate mainly through the employment margin: more people work rather than each individual continuously adjusting hours.

By contrast,in the divisible labor model,

$$TFP \uparrow \Rightarrow \text{demand for labor} \uparrow \Rightarrow W_t \uparrow \Rightarrow N_t \uparrow \Rightarrow 1 - N_t \downarrow$$

$\ln(1 - N_t)$ becomes steeper $\Rightarrow$ the cost of giving up a marginal hour of leisure rises

$$\Rightarrow \quad \text{unwilling to adjust } N_t, \text{ wage moves more} \Rightarrow \text{stiff labor}$$
$$\Rightarrow \quad \text{small hour volatility.} \qquad \text{(each person chooses hours continuously)}$$

Consistency with facts

$$n_t = \frac{H_t}{\bar{L}} = \frac{H_t}{L_t} \cdot \frac{L_t}{\bar{L}} \approx n^*(1 - u_t) = n^* \pi_t,$$

where u_t is the unemployment rate.

$$\frac{H_t}{L_t} = \bar{n}_t \approx n^* \qquad \text{is empirically stable.}$$

*Cyclical movement in total hours comes from (extensive margin) employment, not from hours per worker (intensive margin).

5 Other RBC models

1. CRRA utility (King and Rebelo, 1999):

$$U(C_t, L_t) = \ln C_t + A \frac{(1-N_t)^{1-\chi} - 1}{1-\chi}.$$

   Investment volatility $\uparrow\uparrow$ (almost correct), but underpredicts persistence, over-predicts comovement; consumption & labor are less volatile in model than data.

   change preference curvature and more volatile TFP

2. KPR utility (King, Plosser and Rebelo, 1988):

$$U(C_t, L_t) = \frac{[C_t\, v(L_t)]^{1-\sigma} - 1}{1-\sigma}.$$

   Non-separable preference between C and N. $U_{C,t}$ depends on L_t, U_{L_t} depends on C_t. Relax the too-tight linkages $\Rightarrow$ improve comovement patterns.

3. GHH utility (Greenwood–Hercowitz–Huffman, 1988, AER)

$$U(C_t, N_t) = \frac{1}{1-\sigma}\left[\left(C_t - A\frac{N_t^{1+\psi}}{1+\psi}\right)^{1-\sigma} - 1\right].$$

   Cancel wealth effect, labor supply responds to substitution effect (wage/productivity), $\Rightarrow$ even more volatile N_t.

   Since

$$-U_{N,t} = U_{C,t}W_t \quad \Rightarrow \quad AN_t^{\psi} = W_t.$$

4. Capital utilization:

$$Y_t = Z_t(u_t K_{t-1})^{\alpha} N_t^{1-\alpha},$$
$$K_t = I_t + [1 - \delta(u_t)]K_{t-1}, \qquad \delta(u) > 0$$

   Extra intensive margin for capital $\Rightarrow$ higher investment volatility, higher output volatility.